Alexander Braverman

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Education

Cornell University — Ithaca, NY

B.S. in Computer Science and Operations Research (Double Major) | GPA: 3.89/4.00 — *Expected May 2028*

- **Coursework:** Reinforcement Learning, Machine Learning, Robot Learning, Large-Scale Machine Learning, Data Structures
- **Activities:** Executive Board Member, Artificial Intelligence Undergraduate Club (CUAI) — undergrad ML research group

Research Experience

UCLA Artificial General Intelligence Lab — Los Angeles, CA

Undergraduate Researcher, First Author — *Dec. 2024 – Jul. 2026*

- Built **Test-Time A* Search (TTA*)**, a training-free decoding wrapper that reframes multi-step reasoning as depth-regularized tree search over complete candidate solutions; raised accuracy on 5 open-weight LLMs (1–8B, Llama + Qwen) across 4 math benchmarks by **+10.6–13.8 pts on GSM8K** and **+9.6–27.4 pts on MATH500** over zero-shot CoT
- Outperformed every configuration of Best-of-N, Iterative Self-Refine, and Tree-of-Thoughts at any compute budget swept: **95.4% vs. 92.0%** best baseline on GSM8K (Qwen3-4B) using **4× fewer generated tokens** (6.1k vs. 23.4k per problem)
- Built the evaluation infrastructure for a test-time-compute study: swept **4 methods × 6 budgets × 5 models** over 2,250 problems on vLLM-served models across RTX A6000s, instrumenting per-problem calls, tokens, and wall-clock into accuracy-vs-compute Pareto curves; batched independent self-evaluation samples per node expansion
- Accepted to NeurIPS LAW Workshop 2025; full paper under review at EACL 2027; openreview.net/pdf?id=7qeBpl3WKN; invited to present at Y Combinator ML Research Symposium (2026)

UNC Chapel Hill Zhang Lab — Remote

Undergraduate Researcher, First Author — *Jun. 2026 – Present*

- Leading a new project on **causal credit assignment in POMDPs**: when an agent can't observe the full state of the world, existing methods are not able to tell which action resulted in rewards
- Reformulating credit assignment over the agent's belief rather than the true state, so credit stays correctly assigned when key information stays hidden
- Experiments to be run on MiniGrid and MuJoCo manipulation. Targeting NeurIPS 2027. Advised by Prof. Weitong Zhang

UCLA Artificial General Intelligence Lab — Los Angeles, CA

Research Intern, First Author — *Apr. 2023 – Dec. 2024*

- Built **explanatory prompting**, a method that hands the LLM an informal algorithmic description (DFS) of a task class; cut hallucination rate on a 500-graph DAG-connectivity benchmark from **44.8% to 1.8%** (20.8% → 98.2% accuracy, GPT-4)
- Ablated each prompt component to isolate its contribution: topological-order hint **70.2%**, full algorithmic intuition **95.2%**, one-shot worked example **98.2%**, vs. **55.2%** for the zero-shot CoT baseline

University of Maryland FROOT Lab — College Park, MD

Research Intern, First Author — *Apr. 2022 – Sep. 2023*

- Built a Bloom-filter-based migration telemetry layer in C++ for live Redis shard migration, cutting query loss rate **3.5–3.8×** (11.8% → 3.7%) across 750K-query benchmarks on a **100GB / 50M key-value** store
- Re-implemented DistCache's migration path as a UDP-based transfer protocol streaming 50M 2KB records between source and destination clusters using 10 concurrent sender/receiver threads, with the destination sharded across 5 Redis instances
- Replaced the standard double-query pattern with an **O(1)** destination-side membership probe (integer hash, 125MB bitmap) that forwards unmigrated keys to the source in a single hop, **halving** per-request network cost
- Benchmarked on a 3-node testbed (2× Intel Xeon Gold 5317, 512GB DRAM, 10G NIC) across Zipf skews 0.90/0.95/0.99 and write ratios 0.01–0.05; measured filter error at **<0.043%** of all queries

Publications

- A. Braverman, W. Zhang, Q. Gu. **"Test-Time Scaling for Multistep Reasoning in Small Language Models via A* Search."** *NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models.*
- A. Braverman, W. Zhang, Q. Gu. **"Mitigating Hallucination in Large Language Models with Explanatory Prompting."** *NeurIPS 2024 Workshop on Statistical Foundations of LLMs and Foundation Models.*
- A. Braverman, Z. Liu. **"Toward Fast Query Serving in Key-Value Store Migration with Approximate Telemetry."** *ACM SIGMETRICS Performance Evaluation Review*, 2023.

Projects

remapd github.com/dlin14/remapd | *Python, LangGraph, Claude API, FastAPI, Next.js* — Apr. 2026

- **Finalist, ∼50 teams**: built a redistricting engine (FastAPI + Next.js) that scores congressional maps for all 50 states on population balance, compactness, and voting-rights proxies
- Wrote a simulated-annealing optimizer over county-to-district assignments in NumPy: **+27% median** objective vs. random baselines (18–42% across 5 states), **under 1s** for 700 iterations on 254 counties
- Owned the agent layer: 5-node LangGraph pipeline turning metric vectors into Claude-generated memos, streamed to the browser, with every number verified using a US Census API call

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Python, C++, Java, SQL, Bash | PyTorch, vLLM, Hugging Face Transformers, LangGraph, NumPy | FastAPI, Redis | Git, Linux